THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** an acid	Acid.
2	**(a)** glow dimly	4.5 V is less than the rated 6 V.
3	**(d)** magnesium	Basic oxide, so metal.
4	**(a)** a reversible chemical change	Hydrated and anhydrous forms.
5	**(b)** 40 mL	Proportional.
6	**(a)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
7	**(c)** magnesium	Magnesium is highly reactive.
8	**(b)** formation of a new substance	New substance.
9	**(a)** acidic	Acid rain.
10	**(d)** a conductor	Current passes through it.
11	**(d)** zinc	Acid reacts.
12	**(c)** store in dry place or oil them	Keeps moisture away.
13	**(a)** an acid	An acid neutralises the base.
14	**(a)** glow very bright and may fuse	Exceeds rating.
15	**(a)** sulphur	Non-metal.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
17	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
18	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
19	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
20	Use litmus or turmeric (1); Different colours appear (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
22	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
23	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
24	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
	OR: Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
25	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
	OR: $MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Dissolve equal tablets (1); Add to same volume of acid with indicator (1); Count amount needed to neutralise (1); Same conditions (1); Most effective needs the least acid change (1).	1+1+1+1+1
	OR: All bulbs go off (1); Only one path (1); Test each bulb with a tester (1); Replace the faulty one (1); Use a fresh bulb for check (1).	1+1+1+1+1
27	Colour and lustre (1); Conduction test (1); Magnet for iron (1); Burning gives acidic or basic oxide (1); Reaction with acid (1).	1+1+1+1+1

	OR: Set-ups: dry, no air, tap water, salt water (1); Same nails and period (1); Observe daily (1); Compare rust amounts (1); Salt water most, oiled none (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** copper sulphate	Copper deposits.
2	**(b)** 30 mL	Doubling.
3	**(b)** tap water has dissolved salts	Dissolved ions conduct.
4	**(b)** Fe	Iron.
5	**(a)** lime water test	Test gas.
6	**(a)** it is a base used to relieve acidity	Carbonates react with acids.
7	**(b)** an insulator	Current does not pass.
8	**(a)** copper	Good conductor.
9	**(d)** apply oil and keep dry	Oiling.
10	**(b)** turns almost neutral	They neutralise.
11	**(a)** series	Single path.
12	**(c)** zinc	Zinc.
13	**(b)** carbon dioxide	Lime water test.
14	**(a)** 500 kg	200 x 2.5.
15	**(b)** copper conducts better	Brightness suggests current.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Metals are good conductors (1); They carry current to the appliance (1).	1+1
17	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
18	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
19	Neither acid nor base (1); The effects of acid and base cancel out (1).	1+1
20	Prevent touching live parts (1); Avoid shocks (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
22	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1
23	Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1
24	They are brittle or soft (1); Hammers need hard metal (1); Iron or steel used (1).	1+1+1
	OR: Burning: chemical (1); Light, sound, heat (1); Smoke and gases (1).	1+1+1
25	Add slaked lime (1); Add wood ash (1); Add powdered limestone (1).	1+1+1
	OR: Thin wire heats more (1); Melts first (1); Protects the circuit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Some crops prefer acidic soil (1); Others basic (1); Lime reduces acidity (1); Manure improves soil (1); Test regularly (1).	1+1+1+1+1
	OR: Marks show quality (1); Safer products (1); Fuse rating matches use (1); Do not use higher rating (1); Check before buying (1).	1+1+1+1+1
27	Burn Mg ribbon (1); Dissolve ash in water (1); Burn sulphur and dissolve gas (1); Test both with litmus (1); Compare (1).	1+1+1+1+1
	OR: Burn a candle in a closed jar (1); It goes out (1); Add lime water (1); It turns milky (1); Conclude (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** blue to green	$CuSO_4$ is blue.
2	**(d)** unburnt carbon, a chemical change	New substance.
3	**(a)** a carbonate and an acid	CO2 is released.
4	**(a)** both conduct, copper better	Brightness shows current.
5	**(b)** Q	Reacts fastest.
6	**(c)** to prevent rusting and reduce friction	Both reasons.
7	**(a)** red, purple, blue	Acid to base.
8	**(c)** a single path for the current	The loop is broken.
9	**(b)** less reactive than copper	Copper displaces it.
10	**(c)** the wax changes into new substances such as carbon dioxide and water vapour	Mass is conserved.
11	**(a)** the acid was stronger or more concentrated	Excess acid.
12	**(c)** the nail is coated with paint or rust	Surface layer blocks contact.
13	**(c)** increases	Copper heavier.
14	**(a)** a chemical change	Energy released.
15	**(c)** acid-carbonate reaction	CO2.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They conduct heat away from the hand (1); Heat flows quickly (1).	1+1
17	Takes a long time (1); Small changes each day (1).	1+1
18	A mucus layer protects it (1); The acid is dilute (1).	1+1
19	Nichrome gets very hot when current passes (1); It does not melt or oxidise easily at high temperatures (1).	1+1
20	Gold is very unreactive (1); Silver tarnishes with sulphur (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Colour (1); Density (copper is heavier) (1); Conductivity (1).	1+1+1
22	Chemical (1); New substance slaked lime (1); Heat released (1).	1+1+1
23	Neutralisation releases heat (1); It is exothermic (1); Salt and water form (1).	1+1+1
24	Iron would melt or burn (1); Tungsten has a high melting point (1); Bulb has inert gas (1).	1+1+1
	OR: Soft (1); Reactive with air (1); Surface tarnishes (1).	1+1+1
25	Burning candle (1); Melting is physical (1); Burning vapour is chemical (1).	1+1+1
	OR: Add equal juice (1); Compare the red colour depth (1); More intense means more acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Soil treatment (1); Antacids (1); Acid rain (1); Treating wastes (1); Principle: acid + base gives salt + water (1).	1+1+1+1+1
	OR: Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1).	1+1+1+1+1
27	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
	OR: LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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